Emergencies Involving Indwelling Central Lines

UP-08 | Universal Patient Care

Quick flow

01 Identify the Emergency [EMT / AEMT / Paramedic]

Bleeding • damage • dislodgement • air entry • infusion reaction • infection • occlusion

02 Stop Further Harm [EMT / AEMT / Paramedic]

Stop infusion • close clamp • clamp damaged external tubing proximal to defect • protect open site

03 Suspected Air Embolism? [EMT / AEMT / Paramedic]

Stop infusion, clamp/seal • level supine or recovery position • 100% oxygen • rapid transport

04 Dislodged / Damaged / Bleeding? [EMT / AEMT / Paramedic]

Do not advance, remove, reinsert, or force-flush • direct pressure • sterile or air-occlusive dressing

05 Monitor + Transport [EMT / AEMT / Paramedic]

Cardiac rhythm • SpO_2 • $EtCO_2$ • airway • perfusion • neurologic status • early notification

Overview

- Recognize and stabilize complications involving PICC lines, tunneled central catheters, implanted ports, and dialysis catheters. Prevent hemorrhage, additional air entry, medication exposure, and further catheter damage.
- UP-08 does not authorize routine central-device access, medication administration, or infusion management. Use of an indwelling central device during interfacility transport is governed only by the separate interfacility-transfer policy.

Provider-Level Actions

Higher provider levels include the actions listed for the preceding levels.

EMT

- Identify the device and problem without disconnecting, accessing, flushing, advancing, removing, or manipulating the catheter. Obtain the device and infusion history from the patient, caregiver, sending facility, labels, and pump display when immediately available.
- For line damage, dislodgement, suspected air entry, or an infusion reaction, stop the infusion. Close an existing clamp or clamp intact external tubing between the patient and the damaged segment. Use a purpose-designed line clamp when available and do not apply an unprotected metal instrument directly to the catheter.
- For bleeding at the insertion site, apply direct pressure around the catheter without removing it. If the catheter is completely dislodged, apply direct pressure until hemorrhage is controlled, then cover the site with an air-occlusive dressing. Do not reinsert the catheter.
- If the catheter is partially dislodged, do not advance or remove it. Stabilize the external segment in its found position and cover the site with a sterile dressing.
- For suspected air embolism, stop the infusion, clamp the line, seal any open insertion site, and administer 100% oxygen. Position supine on a level surface; if mental status is impaired but breathing is adequate and no advanced airway is present, use the recovery position on a level surface to protect the airway. Do not routinely use head-down positioning. If cardiac arrest occurs, provide standard CPR supine. Transport rapidly and notify Medical Control and the receiving facility early so hyperbaric consultation can be considered without delaying stabilization.

- Apply cardiac, SpO_2, and $EtCO_2$ monitoring for suspected air embolism. Repeat airway, respiratory, perfusion, neurologic, and complete vital-sign assessments during transport.
- Do not alter an uncomplicated infusion under UP-08. Routine continuation, adjustment, or discontinuation of an interfacility infusion is governed by the separate interfacility-transfer policy.

AEMT

- Perform the immediate complication-control actions above. If medication or fluid remains necessary after the central device is no longer usable, establish peripheral IV or IO access according to UP-06 without delaying transport.
- Treat hypotension or another identified complication under the applicable shock, allergic-reaction, sepsis, or complaint-specific protocol. Do not administer medication or fluid through the affected central device under UP-08.
- If parenteral nutrition is interrupted, obtain a blood glucose promptly and repeat glucose assessment for symptoms of hypoglycemia or during prolonged transport.

Paramedic

- Interpret the cardiac rhythm, manage airway or ventilatory compromise, and treat shock, dysrhythmia, anaphylaxis, seizure, or another identified complication under the applicable protocol.
- For suspected air embolism, continue high-concentration oxygen and cardiac, SpO_2, and waveform $EtCO_2$ monitoring. Do not delay transport for attempts to aspirate air through the catheter.
- Do not access, repair, exchange, advance, remove, or force-flush the catheter under UP-08. Central-device use during an interfacility transfer requires the separate interfacility-transfer policy.

Indications

- Bleeding, damage, breakage, leakage, partial or complete dislodgement, suspected venous air embolism, occlusion, infusion reaction, infection, or other emergency involving an indwelling central device.

Contraindications

- Do not use UP-08 to initiate routine access, administer medication or fluid, or manage an uncomplicated interfacility infusion.
- Do not advance, reinsert, exchange, remove, repair, or force-flush a damaged, dislodged, occluded, painful, leaking, or otherwise abnormal central device.

Assessment | Device and complication assessment

- Identify the device type when possible: PICC, tunneled catheter, implanted port, or dialysis catheter. Determine whether it was accessed before EMS arrival, what is infusing, the ordered rate, when the problem began, and what manipulation occurred immediately beforehand.
- Inspect without probing for external damage, open connections, clamp position, leakage, bleeding, catheter-length change, dressing disruption, erythema, warmth, drainage, swelling, tenderness, and impaired distal circulation.
- Assess for acute dyspnea, chest pain, hypoxemia, cough, hypotension, altered mental status, seizure, or focal neurologic findings suggesting air embolism; assess for fever, rigors, shock, urticaria, bronchospasm, or angioedema suggesting infection or infusion reaction.

Treatment steps

- Stop an infusion associated with catheter damage, dislodgement, suspected air embolism, or infusion reaction. Close an existing clamp or clamp intact external tubing between the patient and the defect.
- Control bleeding with direct pressure. For complete dislodgement, apply an air-occlusive dressing after hemostasis; for partial dislodgement, stabilize the catheter in place and apply a sterile dressing.

- For suspected air embolism, stop the infusion, clamp the line, seal any open insertion site, and administer 100% oxygen. Position supine on a level surface; if mental status is impaired but breathing is adequate and no advanced airway is present, use the recovery position on a level surface to protect the airway. Do not routinely use head-down positioning. If cardiac arrest occurs, provide standard CPR supine. Transport rapidly and notify Medical Control and the receiving facility early so hyperbaric consultation can be considered without delaying stabilization.
- Do not advance, reinsert, remove, repair, or force-flush the catheter. Establish alternate peripheral IV or IO access when ongoing treatment requires vascular access.
- Treat hemorrhage, shock, anaphylaxis, sepsis, respiratory failure, dysrhythmia, seizure, or another complication under the applicable protocol.

Warnings / precautions

- Never force-flush an occluded or resistant central line. Forced flushing may dislodge thrombus, rupture the catheter, or cause infiltration or embolization.
- Do not advance a partially dislodged catheter or reinsert a completely dislodged catheter.
- An open central-line tract may permit air entry even after the catheter is completely removed. Maintain an air-occlusive dressing and monitor for delayed cardiopulmonary or neurologic symptoms.
- Do not delay rapid transport for catheter troubleshooting, repair, or air aspiration.
- A life-sustaining infusion that is not associated with line damage, dislodgement, air entry, or reaction must not be altered under UP-08; follow the separate interfacility-transfer policy and sending/receiving physician orders.

Clinical pearls

- Caregivers and sending-facility staff may provide important device-specific information, but immediate stabilization takes priority when hemorrhage, air entry, or a serious infusion reaction is suspected.
- Preserve the pump, medication container, tubing, labels, disconnected components, and the dislodged catheter when available for transfer with the patient; do not discard them at the scene.
- If parenteral nutrition is stopped, monitor for hypoglycemia, particularly in pediatric patients or during prolonged transport.

Special populations | Pediatrics - younger than 16 years

- Secure damaged or partially dislodged tubing carefully to prevent further movement and use age-appropriate positioning while maintaining airway and respiratory monitoring.
- If parenteral nutrition is interrupted, obtain blood glucose promptly and repeat for symptoms or prolonged transport.

Source and references

Tennessee EMS BLS/ALS State Protocol Guidelines 2024-2025 - Indwelling IV Port Access procedure | Claiborne County EMS UP-08 source PDF - Emergencies Involving Indwelling Central Lines, revised September 1, 2025 | Claiborne County EMS UP-06 IV / IO Access and separate interfacility-transfer policy | Central venous catheter air-embolism literature addressing prevention of further air entry, oxygen, positioning, and occlusive dressing

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